AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I093*

Team Name: *TEEN TITANS*

Domain: *ENERGY , MOBILITY AND SMART INFRASTRUCTURE*

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Presentation Outline

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Problem Statement

->Conventional household electricity monitoring primarily reports total energy consumption, leaving users with limited visibility into room-wise and individual appliance-wise usage.

->This makes it difficult to identify high-consumption appliances, recognize avoidable electricity wastage, and make informed decisions to improve energy efficiency and control electricity costs.

Problem Understanding

- **Who experiences this problem?**
  - Homeowners, tenants, families, building managers, and small businesses seeking better control over electricity consumption and expenses.
- **What causes the problem?**
  - Conventional meters provide aggregated consumption data, while limited integration of room-level and appliance-level monitoring restricts detailed insights.
- **How does it affect users or stakeholders?**
  - Limited visibility into consumption patterns makes it difficult to identify inefficient usage, understand appliance-wise costs, and make informed energy-saving decisions.

Existing Gaps and Challenges

- **Existing Gap 1**
  - Limited integration of room-level and appliance-level monitoring into a unified electricity monitoring framework.
- **Existing Gap 2**
  - Cost, cross-household performance variations, and limited standardized real-world evaluation challenge practical and reliable deployment.
- **Main Unresolved Challenge**
  - Developing an affordable, reliable, and integrated monitoring system that provides meaningful room-wise and appliance-wise consumption insights under real household conditions.

Our Proposed Solution

Core Idea:

- **Room & Appliance-level Monitoring:** Calculates electricity consumption at both room and individual appliance levels.
- **Real-Time Data Tracking:** Uses ESP32, current sensors, and voltage sensors in switch boxes to transmit data in watts.

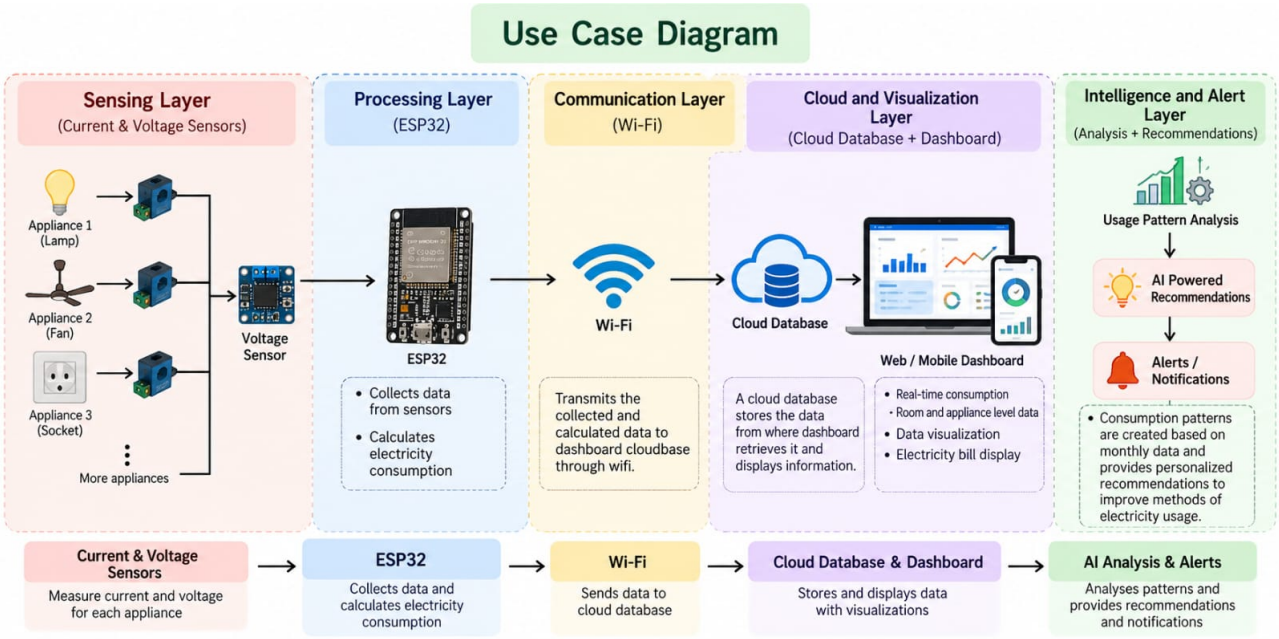
Main Value to the users:

- **AI-Powered Recommendations:** Analyzes usage patterns and visualizations to suggest sustainable habits.

Working at High-Level:

- **Automated Billing:** Generates electricity bills based on power consumed and unit cost.
- **Interactive Dashboard:** Cloud database storage displaying room/appliance consumption, monthly patterns, bills, and data visualizations.
- **Usage Alerts:** Notifies users when consumption crosses predefined limits.

Technical Approach



System architecture showing the interaction between ESP32, sensors, cloud database, and dashboard, illustrating how data flows seamlessly from hardware collection to cloud visualization and AI-driven analysis.

Tech Stack

Frontend: Dashboard

Backend: ESP32

Database: Cloud database

AI/ML, APIs, or Hardware: Current sensors, Voltage sensor, Wi-Fi, AI Powered system

Innovation & Uniqueness

- **What is new or different?**
  - **Room & Appliance-level Visibility:** Provides detailed usage information compared to conventional household-wide meter readings.
  - **Single Microcontroller Architecture:** Uses one microcontroller and one voltage sensor per switch board with separate current sensors, making it cost-efficient.
- **Why is this approach suited to the problem?**
  - **Personalized Consumption Limits:** Users can set custom usage limits beyond which alerts are triggered, encouraging conscious energy saving.
- **What is the key differentiator?**
  - **Automated Electricity Billing:** Links energy consumption data with unit charges to help users easily understand their monthly bills.

Feasibility & Scalability

- **Feasibility:**
  - Uses readily available components such as ESP32, current sensors, voltage sensors, and Wi-Fi.
  - Can be developed and tested initially on a single appliance branch.
  - Can be attached to existing switch boards without much difficulty.
  - Uses affordable components to keep the prototype cost-efficient.

- **Scalability:**
  - Can start by monitoring a single appliance and later expand to multiple appliances and rooms.
  - Applicable in houses, hostels, and institutional buildings.
  - Additional sensing points can be easily added.
  - Future integration with automated load control and renewable energy systems.

- **Risks & Mitigations:**
  - Connecting sensors to existing electrical installations requires proper safety measures.
  - Scaling appliances may increase hardware and cloud storage costs.
  - Wi-Fi stability may affect real-time transmission; resolved using reliable storage and properly rated components.

Cost Estimation

1. Hardware Cost

Component	Cost
ESP32 Wi-Fi microcontroller	478
INA219 current/voltage sensor	247
Smart switch enclosure	249
12V DC power supply	167
PCB, wires and connectors	250
Total Hardware Cost	**1,391**

2. Hosting / Maintenance

- Dashboard/Software: 0
- Cloud hosting & DB: 0–300/mo
- Maintenance: 100–200/mo

3. Estimated Total

Cost Category	Amount
Hardware development	1,391
Software development	0
1st-month hosting	100–500
Estimated Total	**1,491–1,891**

Note: Monthly recurring operating costs range from **100 to 500** depending on cloud database tier and active usage volumes.

Use Cases (1/2)

- **Case 1: Household Electricity Monitoring**
  - Household consumption is tracked room-wise and appliance-wise, updating live to the dashboard where users can set consumption thresholds and receive alerts.
- **Case 2: Industrial Electricity Monitoring**
  - Industry usage is monitored unit-wise and machine-wise on a personal dashboard, recording patterns to easily detect faults when usage deviates from normal behavior.

Use Cases (2/2)

- **Case 3: Room-Wise Monitoring in Hotels and Hostels**
  - Appliance-level power detection per room allows management to accurately track individual resident consumption and update utility bills accordingly for a fair financial system.

Impacts and Benefits

Impacts:

- Improves financial system handling, by providing electricity bill insights.
- Brings in the concept of sustainable electricity usage, by providing consumption rates.
- Integrates IoT, AI-based analysis, and smart switching to make energy management smarter.
- Encourages energy-efficient lifestyles and helps reduce power grid demand.

Benefits:

- Helps track electricity consumption appliance-wise for better insights.
- Analyzes monthly patterns and recommends ways to save energy.
- Notifies users when usage reaches a predefined limit.
- Identifies energy-intensive appliances to manage expenses.
- Displays data through a user-friendly dashboard.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 7	Target 7.3	Affordable and Clean Energy	Promotes energy efficiency by monitoring appliance-level consumption and recommending ways to reduce wastage.
SDG 9	Target 9.4	Industry, Innovation and Infrastructure	Uses IoT technology, smart switches, and AI-based analysis to improve resource efficiency.
SDG 11	Target 11.6	Sustainable Cities and Communities	Encourages households to reduce unnecessary electricity consumption, supporting sustainable communities.
SDG 12	Target 12.8	Responsible Consumption	Increases consumer awareness through real-time monitoring and personalized energy-saving suggestions.
SDG 13	Target 13.3	Climate Action	Promotes energy conservation to help reduce emissions associated with electricity generation.

Individual Member Contribution

Member	Contribution or responsibility
M Dharshan	Dashboard Development & Cloud Integration
Mirnalitha V S	ESP32 Programming & Data Processing
Miruthala Mohan	Hardware Design & Sensor Integration
Dharsshini B A	Data Analysis, Alerts & Documentation

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